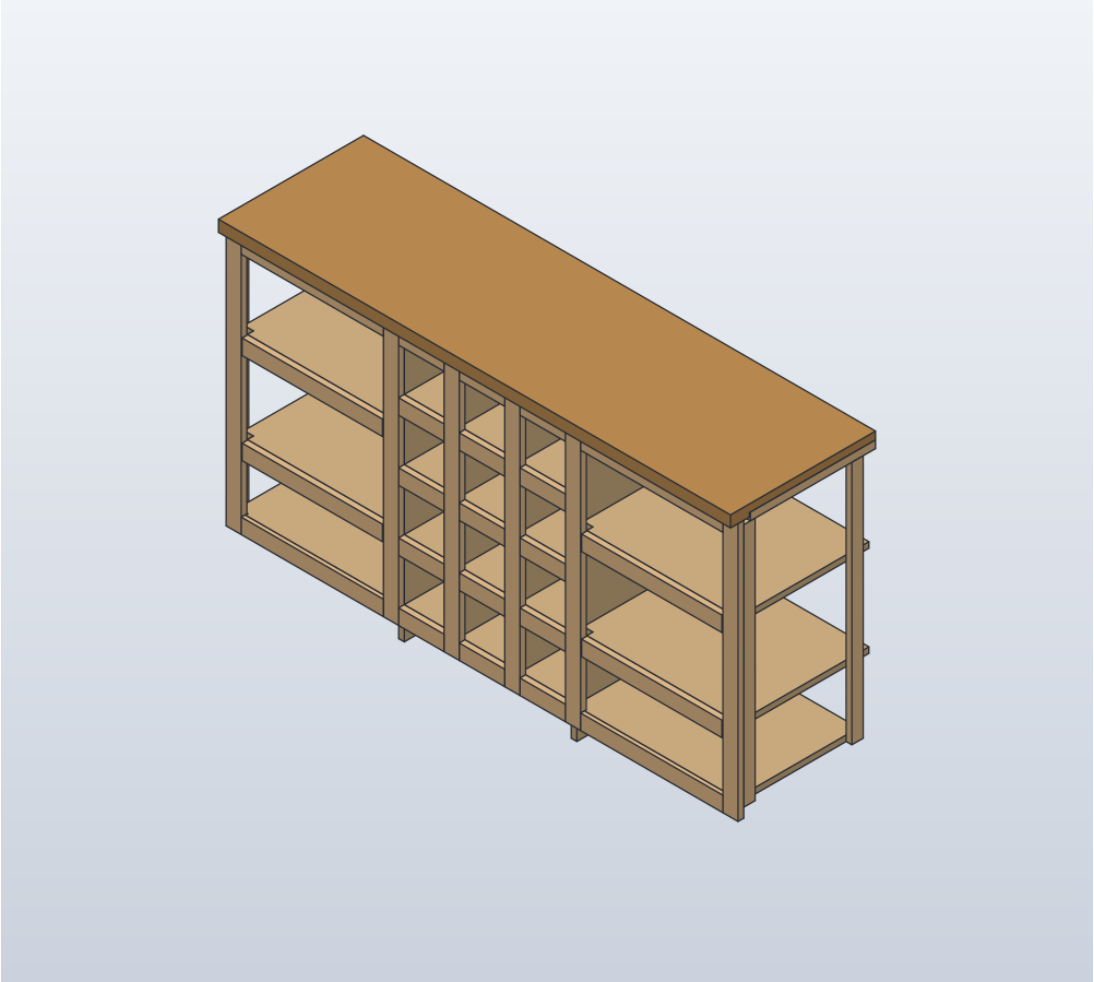
Pantry side shelves

Left and right bays | Nominal opening 67 1/2 in | Field-fit before cutting



Two adjustable shelves and one fixed bottom shelf per bay.

Use four 1/8 in pegs per shelf. The organizer supplies two rows on each outside face.

The wall supplies a front stile and an independent rear pin strip on each side.

A short-depth front pin backer sits behind each visible stile.

The 2 in upper rails hide the lights. Pocket-screw them to the center stiles from behind.

The 1 1/2 in butcher-block top bears on the organizer and hidden side cleats.

Scribe the countertop and fixed bottom shelves. Leave clearance at adjustable shelves.

Cut list

Two shelves per bay shown | Dimensions in inches | Confirm usable stock at the opening

Part IDs	Qty	Material / purpose	Finished L x W x T
TOP-01	1	Countertop butcher block	67 1/2 x 19 1/8 x 1 1/2
CL-L / -R	2	Hidden side cleat hard maple	16 1/2 x 1 x 3/4
ST-L / -R	2	Front stile hard maple	34 x 2 x 3/4
PB-L / -R	2	Front pin backer hard maple	32 1/4 x 1 1/2 x 3/4
PS-L / -R	2	Rear pin strip hard maple	32 1/4 x 1 1/2 x 3/4
TR-L / -R	2	Upper light rail hard maple	18 3/4 x 2 x 3/4
SH-L / -R	4	Adjustable shelf maple plywood	21 1/8 x 17 5/16 x 3/4
FR-L / -R	4	Shelf front apron hard maple	18 1/2 x 2 x 3/4

4/4 stock: two 6 x 96 in boards, 1/8 in kerf.

Board A: rip one 2 1/8 in and two 1 5/8 in rough strips.

2 1/8 in strip: two 33 3/4 in stiles + one 19 3/4 in upper rail.

First 1 5/8 in strip: two 33 1/4 in front pin backers.

Second 1 5/8 in strip: two 33 1/4 in rear pin strips.

Board B: rip one 2 1/8 in and one 1 1/8 in rough strip.

2 1/8 in strip: one 19 3/4 in upper rail + two 19 1/2 in shelf aprons.

1 1/8 in strip: two 17 1/2 in cleats. Rip a 2 1/8 in strip for the two extra aprons.

Plywood: four nominal 21 1/8 x 17 5/16 in blanks. Maintain clearance for pin adjustment.

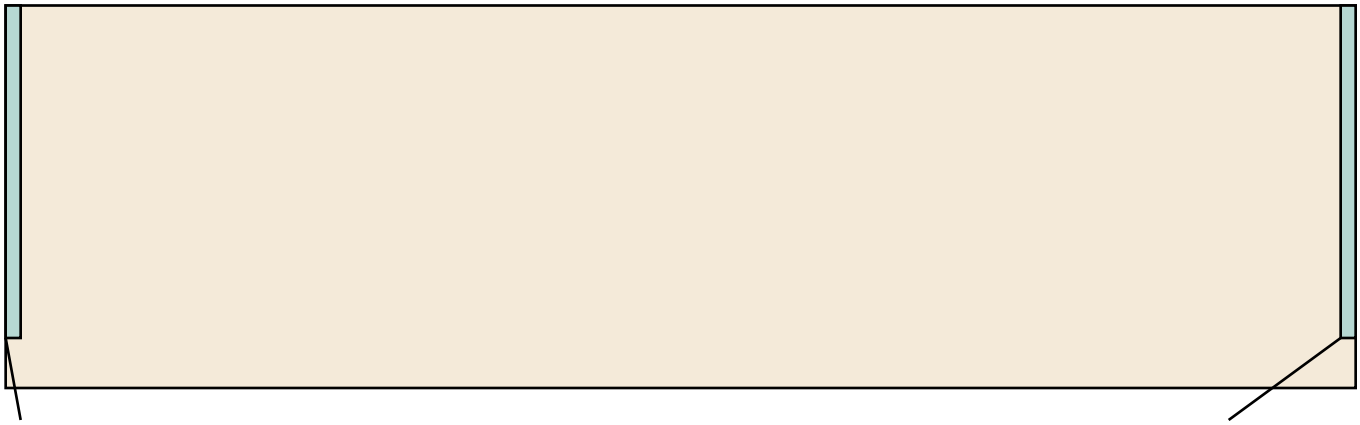
Each 24 x 48 in offcut can yield two shelves, with face grain along the 48 in side.

Counter: template the opening, then transfer the stopped pockets to the underside.

Countertop and cleats

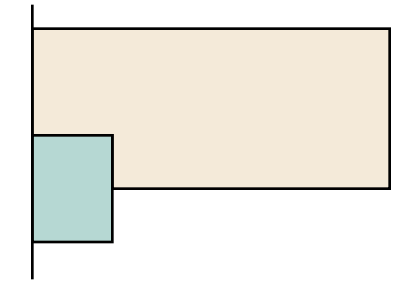
Underside plan and end section | Rear-open stopped edge dados

REAR / CLEAT POCKETS OPEN HERE



FRONT / 1 IN OVERHANG / KEEP 2 1/2 IN OF FULL-THICKNESS NOSE

END SECTION AT CLEAT



WALL

Top: 1 1/2 in thick. Pocket: 3/4 in nominal width x 1/2 in deep.

Cleat: 1 in high x 3/4 in thick. The upper half seats in the pocket.

Fasten cleat into framing. Keep the top free to move front to back.

Grain runs left to right. Allow 1/8 in at the stopped shoulder as drawn.

Fasten cleats before setting the top. Recess heads so the top can slide on the cleats.

Use figure-8s or slots at accessible bearing points. Keep the cleat pocket a slip fit.

Check the top can lower onto both cleats in the real opening before final routing.

Adjustable shelves and lights

Use the center face frame as the front datum | Left and right are field-fit

1. Install the organizer with its 1 1/2 in x 3 1/2 in baseboard reliefs.
2. Fasten each front stile and pin backer, then locate the independent rear pin strip.
3. Dry-fit the low shelf and its 2 in apron. Lift and remove it past the upper trim.
4. Transfer one level datum to all four peg rows. Match peg bearing heights.
5. Install upper decorative rails, then mount channels to their rear faces.

Peg rows (from front face of center organizer)

Front wall-side backer: 1 7/8 in. Rear strip center: left 12 3/4 in, right 15 3/4 in.

Organizer-side front: 1 7/8 in. Organizer-side rear: 15 3/4 in.

Wall-side rear positions are provisional. Keep the left strip clear of plugs and cords.

Use a common drilling template. A 1 in pitch is shown as a planning choice.

Start the useful pin range near 6 in above floor. Set the upper limit after dry-fit.

Check pin projection across the scribe gap and use a depth stop sized to actual pegs.

Organizer side residual thickness can be 1/2 in at internal recesses. Avoid breakthrough.

Rectangular lighting channels

Mount channels to the rear faces of the middle shelf aprons, aimed toward the rear.

Mount fixed channels to the rear faces of the rails directly below the countertop.

Channel blanks: two up to 18 1/2 in at the shelves and two up to 18 3/4 in above.

Leave extra end clearance for caps and leads. Confirm actual extrusion and clip dimensions.

The CAD envelope is 3/8 in front-to-back x 5/8 in high. Check it before machining.

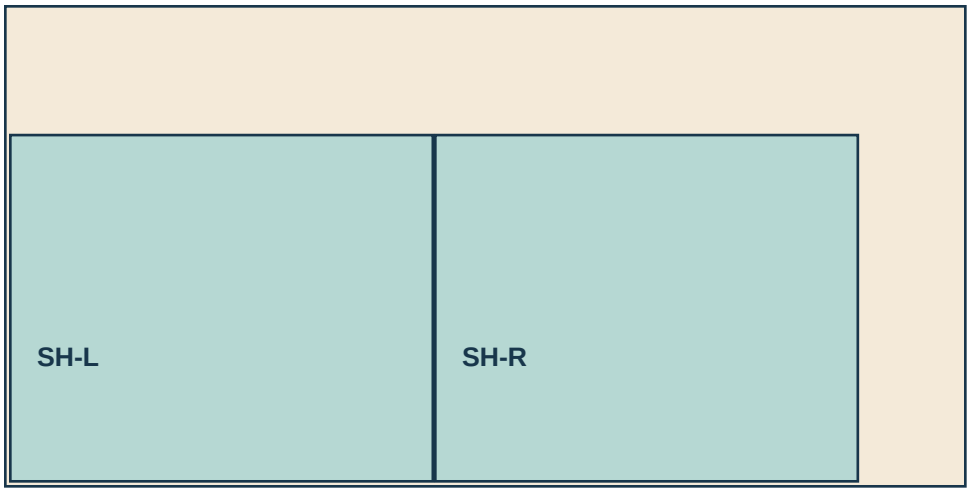
Use a removable low-voltage connector and strain-relieved service loop on each shelf.

Allow the wire to move through the shelf adjustment range and keep it clear of peg rows.

Shelf blanks and peg layout

Top views | Shelf front is at the bottom | Peg positions shown at the nominal fit

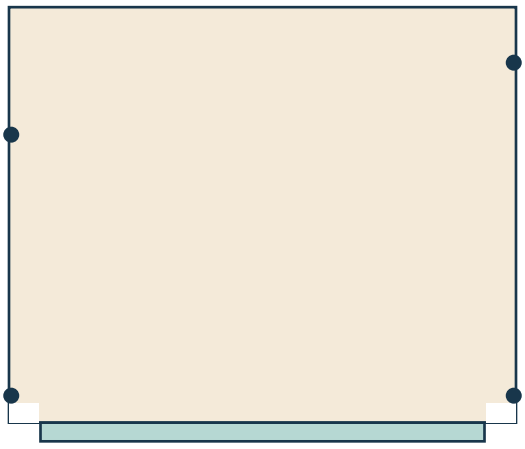
3/4 IN MAPLE PLYWOOD / REPEAT ON TWO 24 X 48 IN OFFCUTS



Trim 1/4 in per edge. Rip 17 5/16 in, then crosscut two 21 1/8 in blanks.

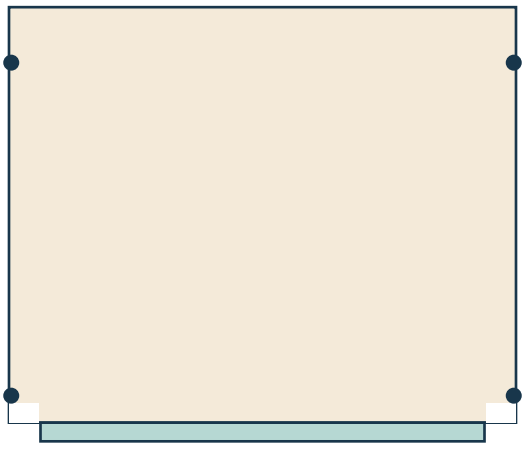
Keep the 1/8 in kerf on the waste side. Keep the adjustable shelf clearance.

WALL / ORGANIZER



SH-L / front apron shown in teal

ORGANIZER / WALL



SH-R / front apron shown in teal

Shelf width = carcass to wall-side pin face minus 1/8 in total.

Apron width = clear opening between stiles minus 1/4 in total.

Front reliefs: 1 1/4 in wide x 13/16 in deep at both corners. Mirror left and right.

Fixed bottom shelves

One per bay | Install after the face frame | Field-fit at the opening

Finished blank sizes before fitting

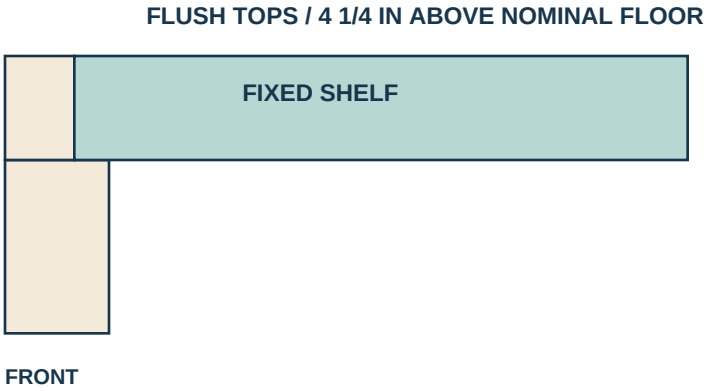
BR-L / BR-R: 18 3/4 x 2 x 3/4 in maple.

BS-L / BS-R: 21 1/4 x 16 1/8 x 3/4 in plywood.

Allow one additional 24 x 48 in offcut for the two bottom blanks.

Fit each rail tight between the stiles and pocket-screw from behind.

Section through bottom rail and shelf



Rabbet: 1/4 in into rear face, down by actual shelf thickness.

Dry-fit shelf thickness. The open rabbet lets the shelf drop in from above.

Pocket-screw the rails before seating the shelves. Keep screw bores below the rabbet.

Scribe each fixed shelf to the actual opening and baseboard.

Cut the two front shoulders so only the tongue between the stiles enters the rabbet.

Mirror the wall-side notch: left front corner on BS-L, right front corner on BS-R.

Stile relief: 1 5/16 in wide x 5/16 in back from the tongue front, through thickness.

This adds 1/16 in at the stile. Transfer the actual stile outline before cutting.

Add rear and side bearing cleats as needed in the field. The rail supports the front.

Secure the bottom shelf after fitting. The shelf pins serve the adjustable shelves.

PLAN AT FRONT EDGE / BS-L SHOWN, MIRROR FOR BS-R



Wall-side relief clears the stile.

Opposite shoulder clears the center stile.

Tongue seats in the rail rabbet